

Instawards Statement of Work (SOW) 30-Day Scoped Engagement

1. Project & Team Information

Project Name:	Keel
Builder / Team Name:	Ciganytry
Primary Contact:	Rafli Ahmad Denistri - radenistri@gmail.com
Ambassador Chapter:	Indonesia
Ambassador Chapter Lead:	Kenny Rivaldi
Date Submitted:	August 2026
Suggested Sprint Start Date:	August, 2026

2. Instawards Overview & Intent

2.1 Instawards Purpose (for Builder Context)

Instawards are designed to support short, clearly scoped, execution-focused work that helps a project make tangible progress toward building on Stellar. Instawards are meant to fund specific, achievable outcomes that can be completed and demonstrated within 30 days or less.

This SOW represents a shared commitment between the Builder and the Ambassador Chapter Lead on what will be delivered, why it matters, and how success will be verified.

3. Problem Statement & Objective

Problem Being Addressed	What specific problem, gap, or blocker is this Instaward intended to solve?	On May 20, 2026, an attacker increased the price of USTRY by 100 times through a thin-liquidity oracle feed, then borrowed \$61 million in XLM against incorrectly valued collateral on Blend. Blockaid and Stellar validators coordinated the quarantine of approximately 48 million XLM through the CAP-77 Quorum Freeze, containing \$10 million in losses. The oracle reported the observed price. The missing control was an independent way to determine whether that price was supported by enough executable market depth to be trusted as collateral. Comparable structural risk remains on-chain. For example, USDY was reported at approximately \$109.58 million in supply, 684 holders, and only \$191 in 24-hour volume as of February 26, 2026. These
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		figures will be verified again during Week 1. Stellar already has threat-detection services, including OctoPos RMS, Blockaid, Hypernative, and Range, that identify anomalies during or after an attack. What is missing is a preventive risk-parameter layer that measures effective liquidity depth per asset before losses occur. Related functions are performed by Gauntlet and Chaos Labs in other ecosystems, but neither currently serves Stellar. This gap is increasingly important following publication of the Stellar DeFi Security Standards and growth in on-chain real-world asset value.
Objective of This Instaward	In one or two sentences, what will be true at the end of 30 days if this Instaward is successful?	At the end of 30 days, Keel will provide a proof-of-concept Liquidity Risk Engine that tests whether effective liquidity depth calculated from SDEX orderbooks and AMM pool reserves would have placed USTRY below a safe threshold before the Blend exploit. The sprint will deliver a reproducible historical backtest using the permanently recorded May 2026 ledger data, a liquidity-depth calculation engine, a public read-only API, an asset-risk dashboard, and an open incident report. These outputs will form the technical foundation for an SCF Build Award application and provide concrete material for validating willingness to pay with protocols and institutional vault operators.

4. Scope of Work (30-Day Deliverables)

Important guidance: This scope must be achievable within **30 calendar days**. If the work feels larger, it should be reduced or split into more achievable phases.

4.1 In-Scope Deliverables

Deliverable	Description (What will be built or produced?)	Why this matters
Deliverable 1	Liquidity Depth Engine. A backend service that reads SDEX orderbooks, AMM liquidity-pool reserves, and Stellar trustline distribution. For each asset, it calculates effective depth at +/-2%, +/-5%, and +/-10%, holder concentration, volume-to-supply ratio, time	These values do not exist on-ledger until they are calculated. Existing monitoring tools observe prices or active threats, but do not quantify

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	since the last genuine trade, and a recommended maximum safe collateral size. It also supports historical replay at prior ledger states. The methodology will be documented and reproducible.	whether a price is supported by executable market depth. This engine measures the risk parameter that was missing in the Blend incident.
Deliverable 2	Public Risk API and Blend Backtest Report. A public read-only HTTP API will expose current depth metrics through GET /asset/{code}:{issuer}/depth and historical metrics through the ledger query parameter. An open, reproducible report will reconstruct USTRY liquidity metrics across May 2026 and identify when the unsafe threshold was crossed relative to the exploit.	The backtest provides an independently verifiable test against a known outcome. The API allows protocols to consume the resulting metrics with minimal integration and supports protocol adoption and commercial validation.
Deliverable 3	Asset Risk Dashboard and supporting materials. A public web dashboard will show monitored Stellar assets, depth curves, risk scores, recommended maximum safe collateral size, and clear flags for high-risk profiles. Each asset will have a detail view with historical trends. The demonstration set will contain at least 50 active Stellar assets. Final materials include methodology documentation and a 3 to 5 minute demo video.	The dashboard makes hidden collateral risk visible to lending protocols, vault curators, and real-world asset issuers. It also provides a reviewable demonstration artifact for the Ambassador and for a future SCF Build Award application.
Out-of-Scope (Explicitly Not Included)		
	• Declarative baselines and reconciliation loops (C1-C3), which are deferred to the SCF Build phase. • Oracle-anomaly and attack-pattern detection (C5 and C9), which existing providers already serve. • Captive-core ingestion. The proof of concept uses public Horizon and RPC endpoints. • Alert delivery through webhook, Slack, or PagerDuty. • Asset-issuer risk profiling (C7). • Signing or submitting transactions. Keel remains permanently read-only. • Publishing a price feed as an oracle, supporting non-Stellar chains, or providing a production mainnet SLA.	

4.2 Deliverable-Aligned Budget Request

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Requested Budget Amount	Rationale for Budget Request
\$5,000 278 combined hours × \$18/hour ≈ 139 hours per builder over 30 calendar days ≈ 4.6 hours per day per builder	The requested amount covers approximately 278 combined engineering hours contributed by two builders during the 30-day Instawards execution period. The estimate is based on a blended engineering rate of \$18/hour, with effort allocated according to the implementation complexity of each deliverable. Each estimate includes the engineering work required for implementation, testing, documentation, integration, and final validation. Deliverable 1: Liquidity Depth Engine and Historical Replay: Implementation of SDEX orderbook and AMM liquidity ingestion, liquidity-depth calculations, historical replay capability, methodology validation, recommended collateral sizing logic, and supporting technical documentation. 126 hours = \$2,268 Deliverable 2: Public API and Blend Backtest Report: Implementation of the public read-only API, historical reconstruction of the Blend incident, reproducible backtest report, API documentation, and validation. 82 hours = \$1,476 Deliverable 3: Asset Risk Dashboard: Implementation of the public demonstration dashboard, visualization of liquidity metrics and risk scores, frontend integration with the API, deployment, demonstration video, and preparation of final review materials. 70 hours = \$1,260 Total: Approximately 278 engineering hours = \$5,000

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5. 30-Day Execution Plan & Timeline

5.1 Weekly Breakdown

Week	Planned Work	Expected Output
Week 1	• Verify current stablecoin data. • Build readers for SDEX orderbooks and AMM pool reserves. • Define and document the liquidity-depth methodology. • Cross-check output against Horizon on sample ledgers.	• Working orderbook and pool ingestion. • Documented methodology. • Cross-validation passed on at least 50 sample ledgers.
Week 2	• Implement depth calculations at +/-2%, +/-5%, and +/-10%. • Implement holder concentration and safe-collateral calculations. • Build historical replay. • Run the engine across at least 50 active Stellar assets.	• Live and historical calculations working. • Initial results for at least 50 assets.
Week 3	• Run the Blend incident backtest on May 2026 ledgers. • Draft and verify the findings. • Build and document the public read-only API.	• Verified backtest report. • Live and documented API.
Week 4	• Build the asset-risk dashboard. • Complete testing and fixes. • Finalize documentation and a 3 to 5 minute demo video. • Publish the backtest report openly.	• Live dashboard. • Demo video. • All final artifacts publicly available.

6. Evidence of Completion (Required)

Important guidance: Evidence should be clear, verifiable, and easy to review by the Ambassador Chapter Lead **with minimal technical expertise**.

6.1 Planned Evidence to Be Submitted

Deliverable	Evidence Type (link, repo, demo, screenshot, document, etc.)	Description
Deliverable 1	Public repository link and methodology document.	Open-source repository containing the Liquidity Depth Engine, reproducible methodology, and cross-validation results against Horizon on sample ledgers.

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Deliverable 2	Live API URL, open backtest report, raw data, and calculation code.	Public API and documentation plus a written Blend incident backtest report that can be independently reproduced and verified.
Deliverable 3	Live dashboard URL and demo recording.	Public dashboard covering at least 50 Stellar assets and a 3 to 5 minute workflow demonstration.

6.2 Evidence Verification Checklist (For Ambassador Use)

For each deliverable, the Ambassador Chapter Lead will assess whether evidence is present and sufficient.

Deliverable	Evidence Present	Evidence Partial	Evidence Missing	Comments
Deliverable 1: Liquidity Depth Engine	☐	☐	☐	
Deliverable 2: Public API and Backtest Report	☐	☐	☐	
Deliverable 3: Asset Risk Dashboard	☐	☐	☐	

7. Next-Step Alignment

7.1 Anticipated Next Step After Completion

After this Instaward, the most likely next step is:

- ☒ Apply to SCF Build Award
- ☐ Continue development independently
- ☐ Apply for a follow-on Instaward (if eligible)
- ☐ Seek other ecosystem support
- ☐ Other:

8. Instawards Constraints Acknowledgement

By submitting this SOW, the Builder acknowledges:

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- ☒ This scope will be completed within **30 days or less**.
- ☒ Instawards support execution, not open-ended exploration.
- ☒ A project may receive **no more than two follow-on Instawards**.
- ☒ Each Instaward is capped at **\$5,000**.
- ☒ Total Instawards funding may not exceed **\$15,000**.

9. Submission Confirmation

Once finalized, this Statement of Work will be submitted by the Ambassador Chapter Lead via the Instawards Airtable submission form for review and approval.

Appendix - Prior-Art Verification Notes

The SOW scope was filtered through a structured review of the Stellar ecosystem directory as of July 31, 2026. Capabilities already served by established providers are intentionally excluded so the sprint remains focused on the identified gap.

Capability Already Available	Existing Providers or Projects
Oracle-anomaly detection	OctoPos RMS (crediolabs/Untangled, live June 2026); Stellar Oracle Shield (SCF #43)
Real-time attack-pattern detection	OctoPos, Blockaid, Hypernative, Range
Formal pre-deployment verification	Certora, Inference, CertiK
KYT, AML, and transaction compliance	Scorechain, Merkle Science

Identified gap

No Stellar project was identified that publishes per-asset effective liquidity depth at +/-2%, +/-5%, and +/-10% together with a recommended maximum safe collateral size. Gauntlet and Chaos Labs perform related functions elsewhere but are not currently present on Stellar.

Commercial risk acknowledged

Their absence may indicate that Stellar is not yet large enough for this business model. The \$5,000 scope therefore prioritizes a limited and independently checkable validation artifact, the Blend backtest, before any full-scale SCF Build commitment.

All dated figures in this SOW originate from the ecosystem snapshot dated July 31, 2026 and will be independently verified again where noted.